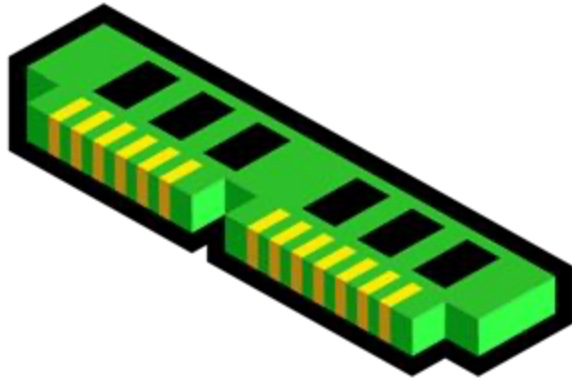


# Neuromechanism of long-term memory retrieval

11 November 2024

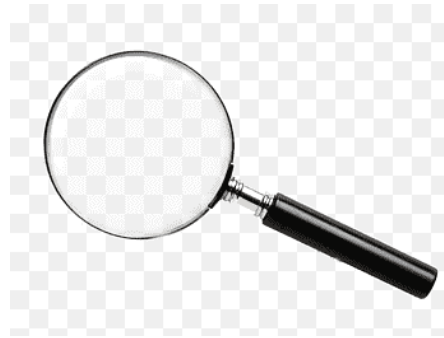
Mr. Poopa Kaewbuapan

# What is Memory and Attention?



Short-term  
Memory (Working)

Active  
representation  
of information



Attention

Selective  
Filter

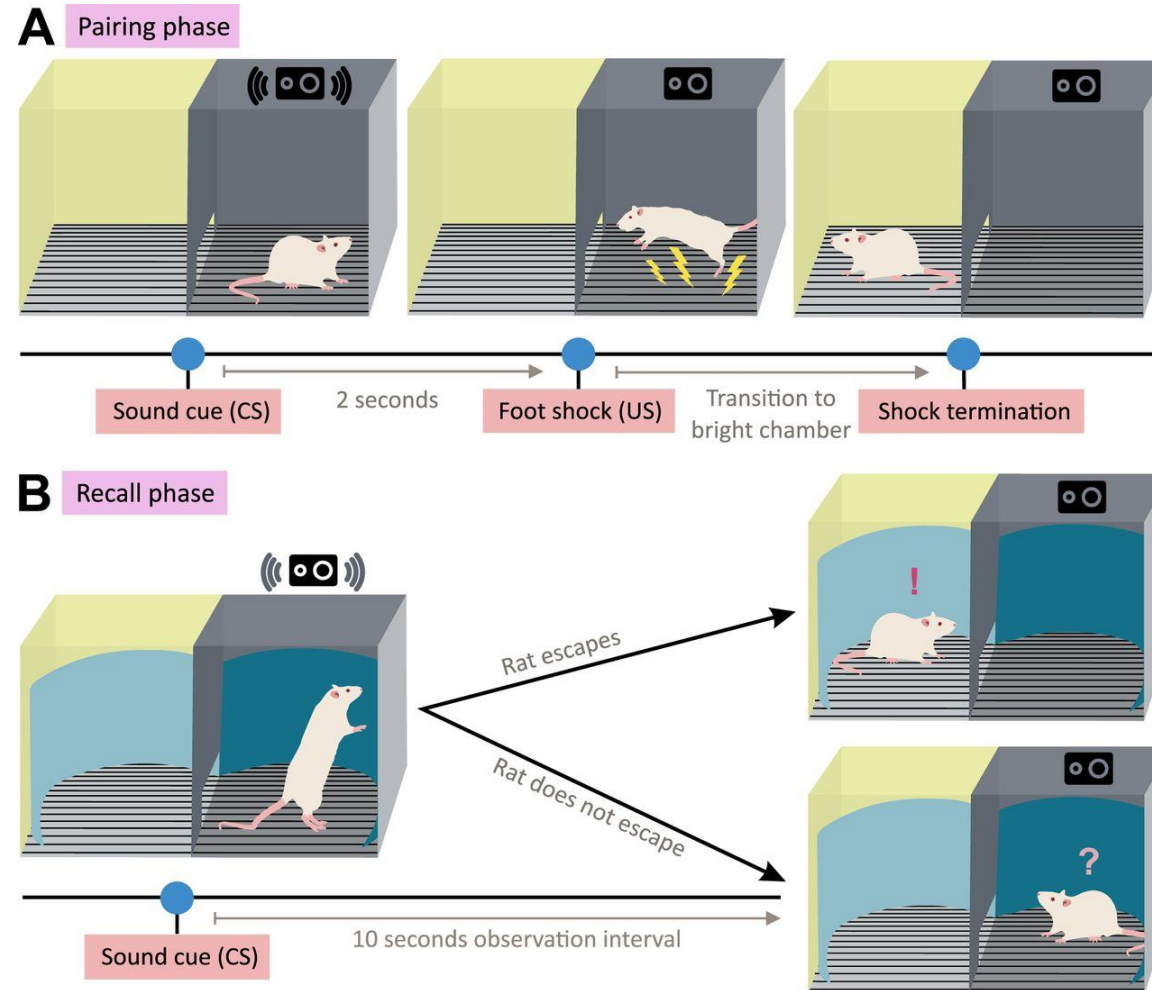


Long-term  
Memory

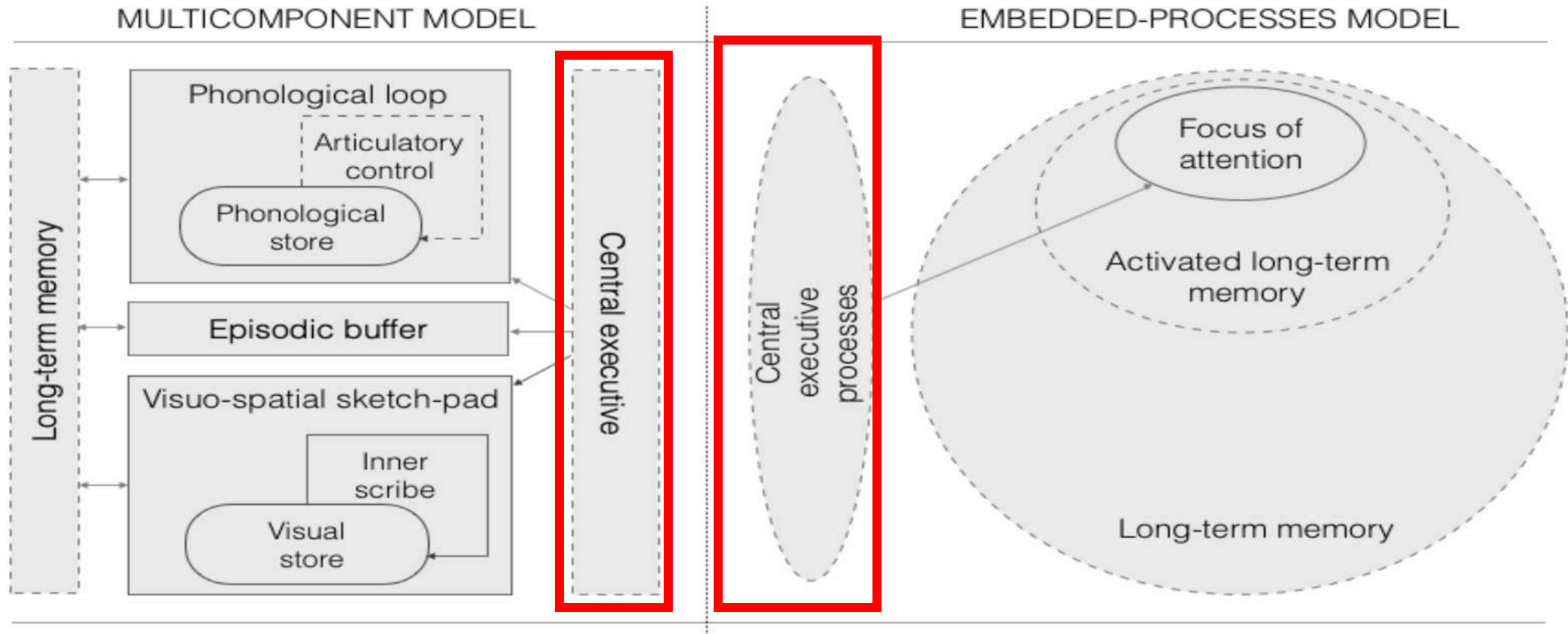
Storage for later  
re-use

# Why LTM retrieval is important?

- To live a (long) life!



# What could limit LTM retrieval?



Baddeley, Logie 2000  
Cowan 1999

Global attentional  
control!

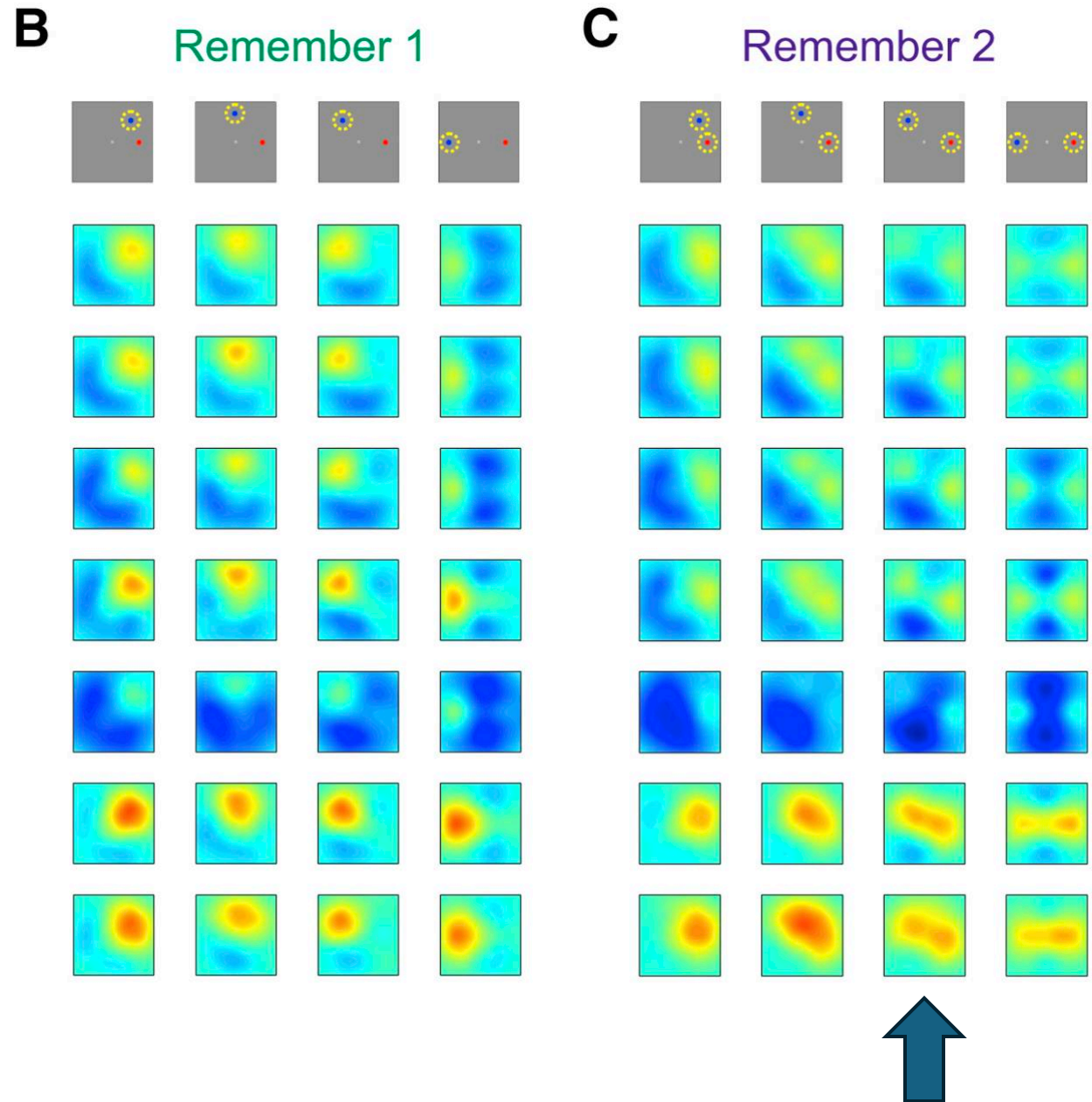
# WM and attention

fMRI agrees!

- Reduced Fidelity on Remember 2 (attention spreading out)

Reconstructions of Information in Visual Spatial Working Memory Degrade with Memory Load

Sprague, Ester, Serences, 2018



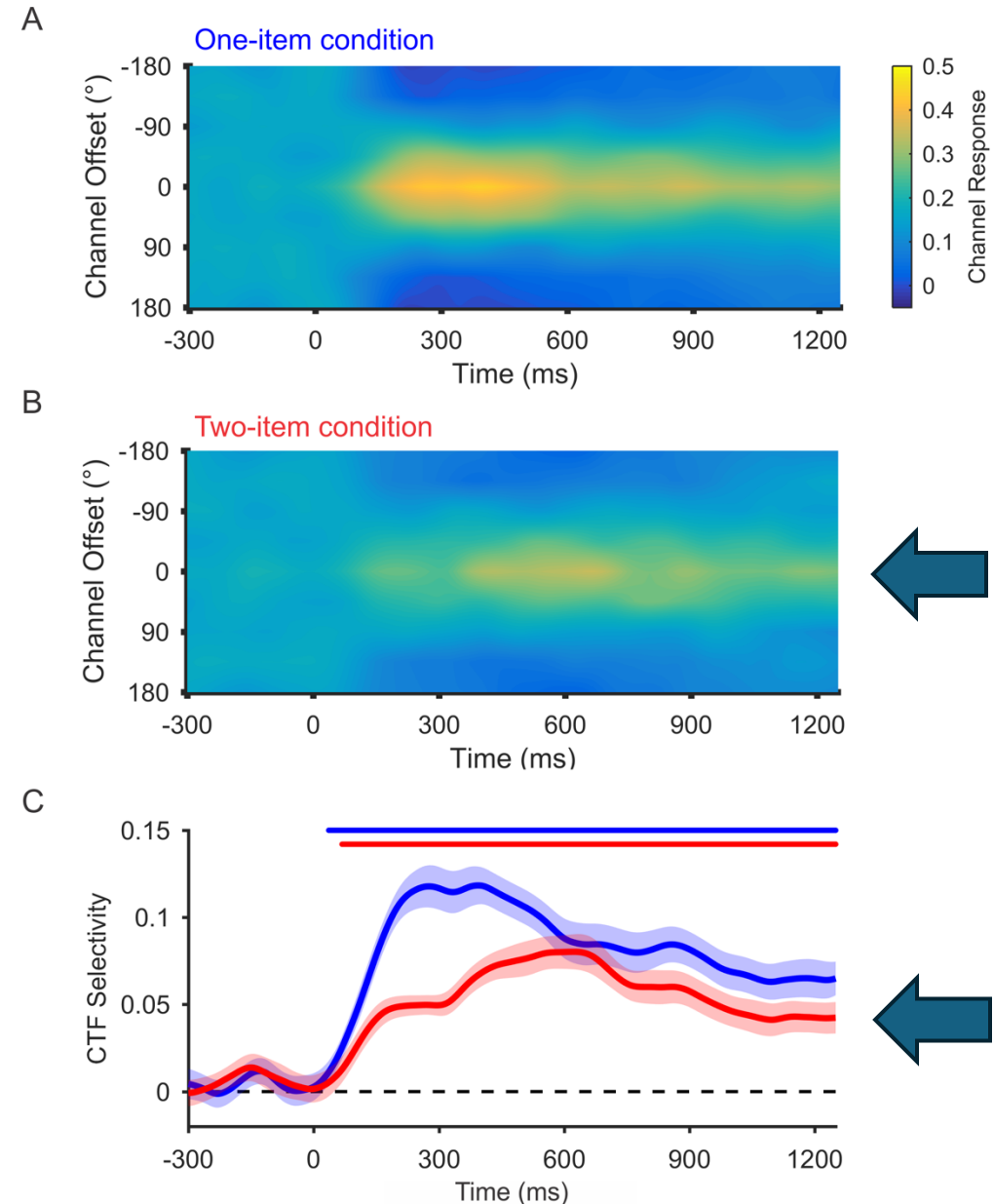
# WM and attention

EEG agrees!

- Reduced location selectivity on two-item condition (less attention gain)

**Item-specific delay activity demonstrates concurrent storage of multiple active neural representations in working memory**

Sutterer et al., 2019



# Objectives

Would attention really modulates LTM retrieval like WM?

- 1. Study how attention interact with the content from LTM retrieval
- 2. Identify neuromechanism that underlying LTM retrieval process
  - ⇒ Exp 1: Confirms that WM neuromechanism is attention
  - ⇒ Exp 2: See if LTM retrieval and WM shares attention

# Data Collection

- $n = 17$
- Convenience sampling, Adult Thai (KMUTT undergraduates) with age range from 18-35 years old

IRB Approved

Data Collection Period: 15/08/2023 - 04/10/2023 (Completed)

Participant Compensation\*: 300 bahts per day session

(Exp 1 + 3 days Exp 2)

\*Funded by Neuroscience Center for Research and Innovation (Nx), Learning Institute, KMUTT

# Experiment Procedure

Total of 4 days -> 1 + 3 *consecutive* days

Day 1: Experiment 1 (2 hrs)

- Measuring WM precision

Day 2-4: Experiment 2 (3 hrs/day)

consists of 4 repetitions of 3 blocks interleaving in order

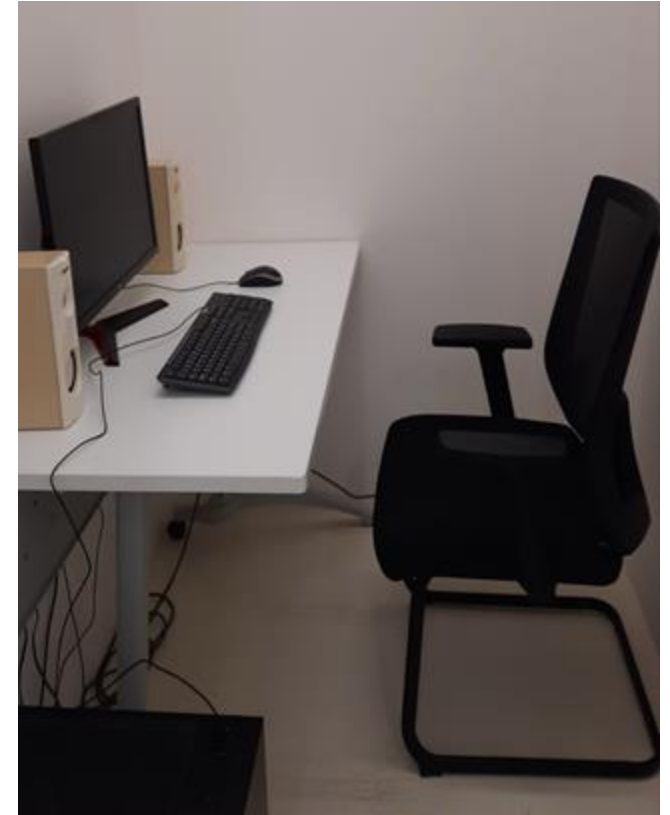
- Study (Encoding into the LTM)
- Retrieval
- Retrieval individual (Measuring LTM baseline)

Collected data

- Directional response error [-180° left 180° right]

=> Degrees deviated from correct location on a circular response ring

- Reaction time (seconds)



Data acquisition setting

# 4-days span experiment

Day 1

Day 2-4

Exp 1 WM

12 repetitions x 7.5 mins

Total 1.5 hrs

Exp 2 – Study block

10 mins

Exp 3 – Retrieval block

10 mins

Exp 4 – Retrieval  
individual block

10 mins

Interleaving for 4 times per day

# Demographics

- Male 11: 64.70%, Female 6: 35.30%
- All age 18-25
- KMUTT undergraduates, Thai

# Experiment 1: Working Memory

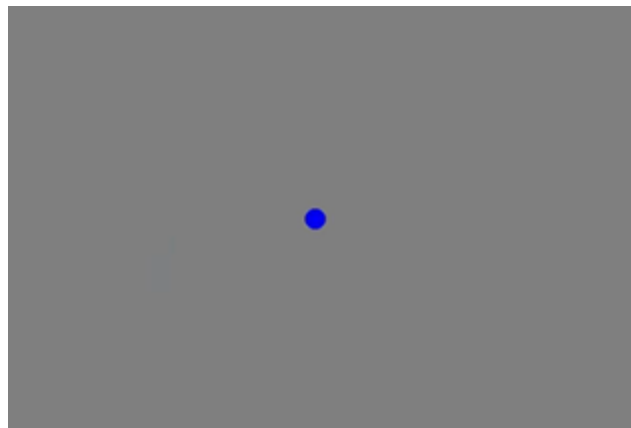
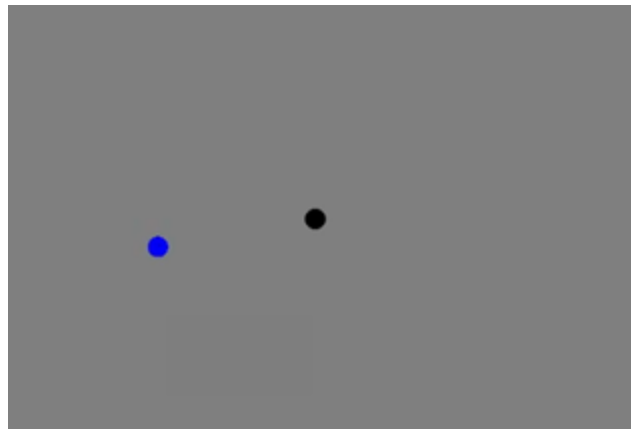
“Confirming that increasing **attentional demand** would reduce WM precision”

Adapted task from Ester et al., 2018.

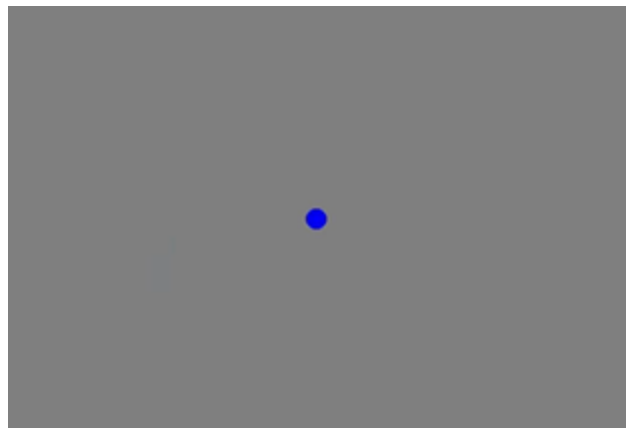
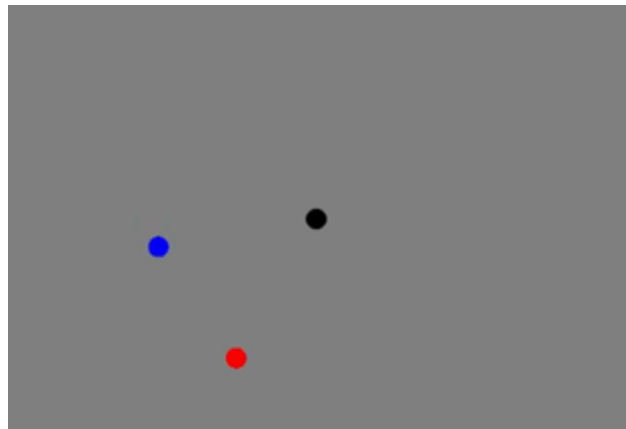
Participant respond remembered spatial location after short delay (need to use WM) according to a color cue.

3 conditions: 1 stimulus, 2 stimulus Focused (increase cognitive load), 2 stimulus Divided (increase both cognitive load and attentional demand)

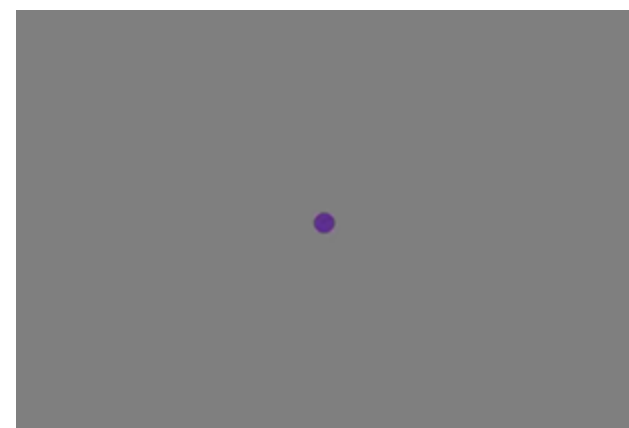
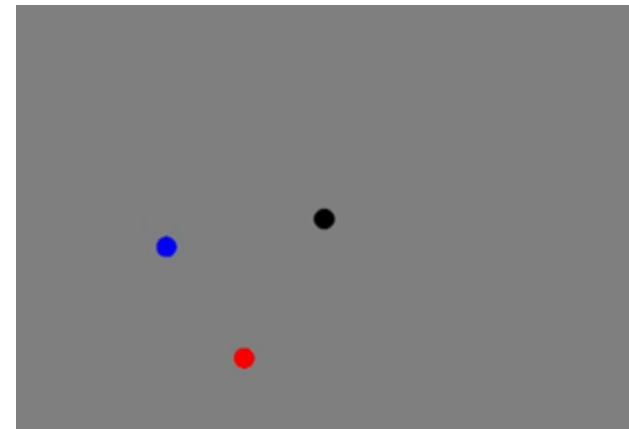
1 stimulus



2 stimulus Focused



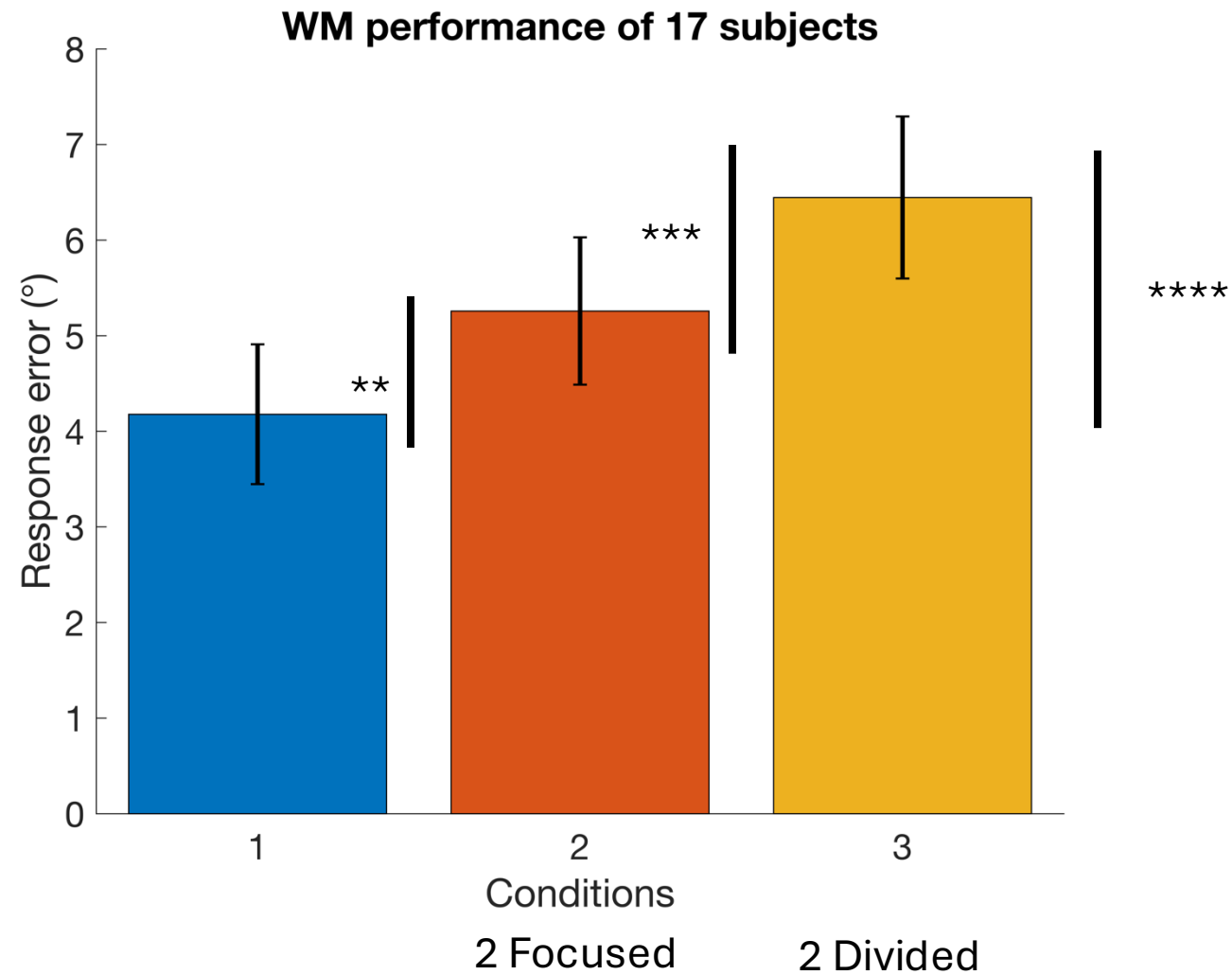
2 stimulus Divided



\*purple cue  
as a distractor

Time

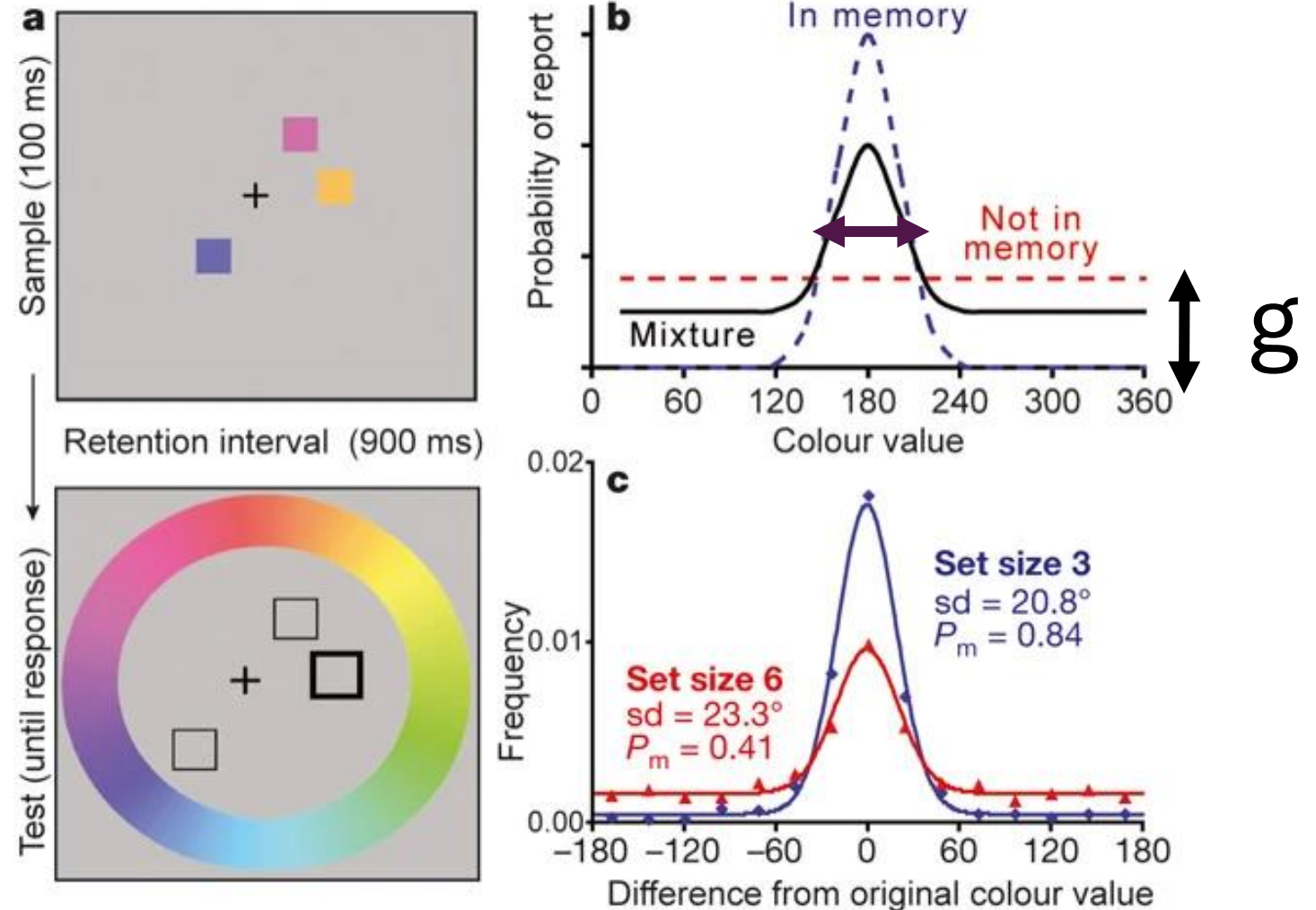
# Exp 1 Result



# Mixture Model (Zhang, Luck, 2008)

Add uniform distribution under the Gaussian curve in case of guessing (target not in memory)

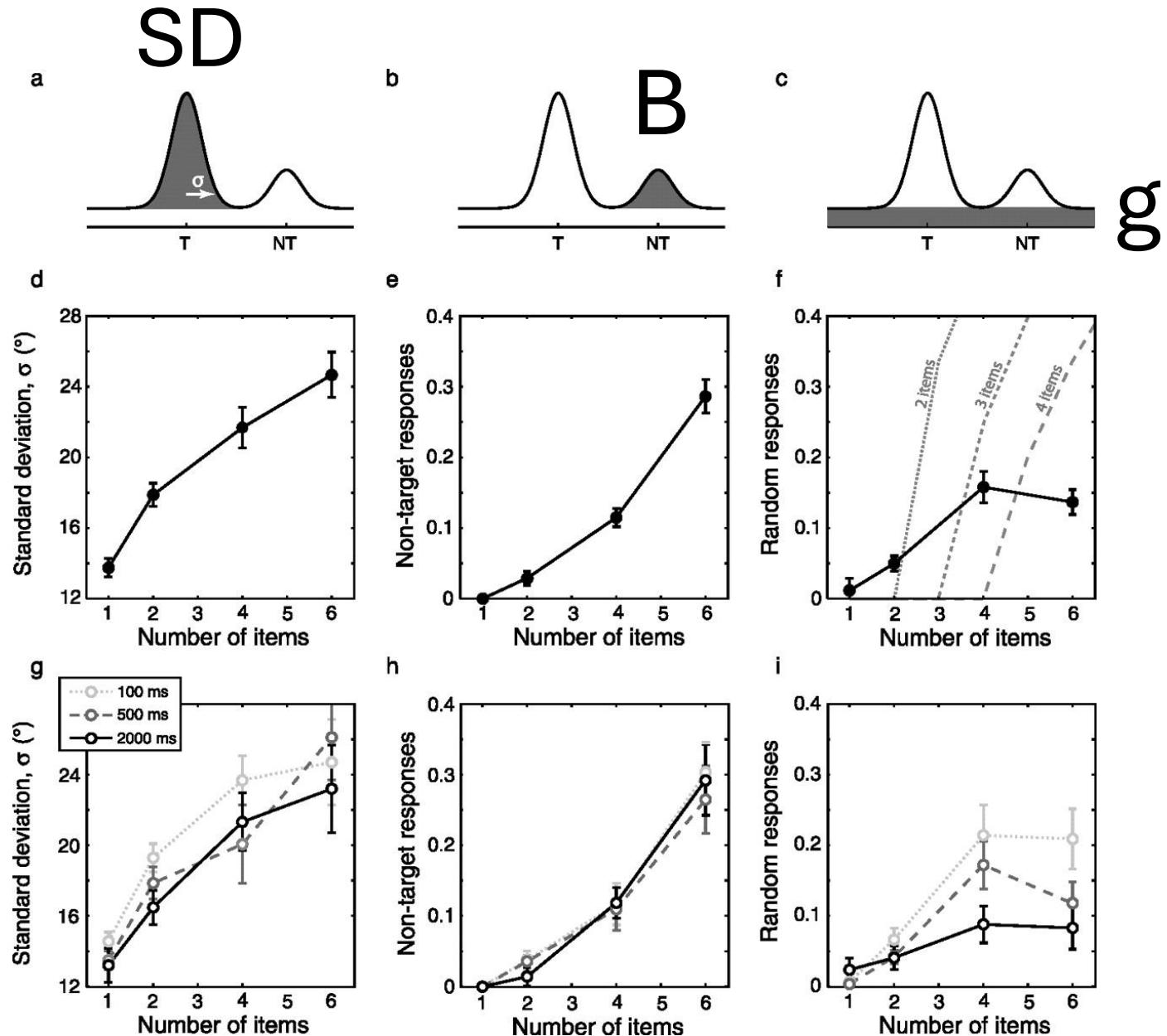
1.  $g$  (Guessing Rate – Probability of choosing randomly)
2. SD (Memory Precision)



# Swap Model (Bays, 2009)

Extended from Mixture Model  
with new parameter  
accounted for distractor

1. B (Probability of choosing non-target location)
2. g (Guessing Rate – Probability of choosing randomly)
3. SD (Memory Precision)



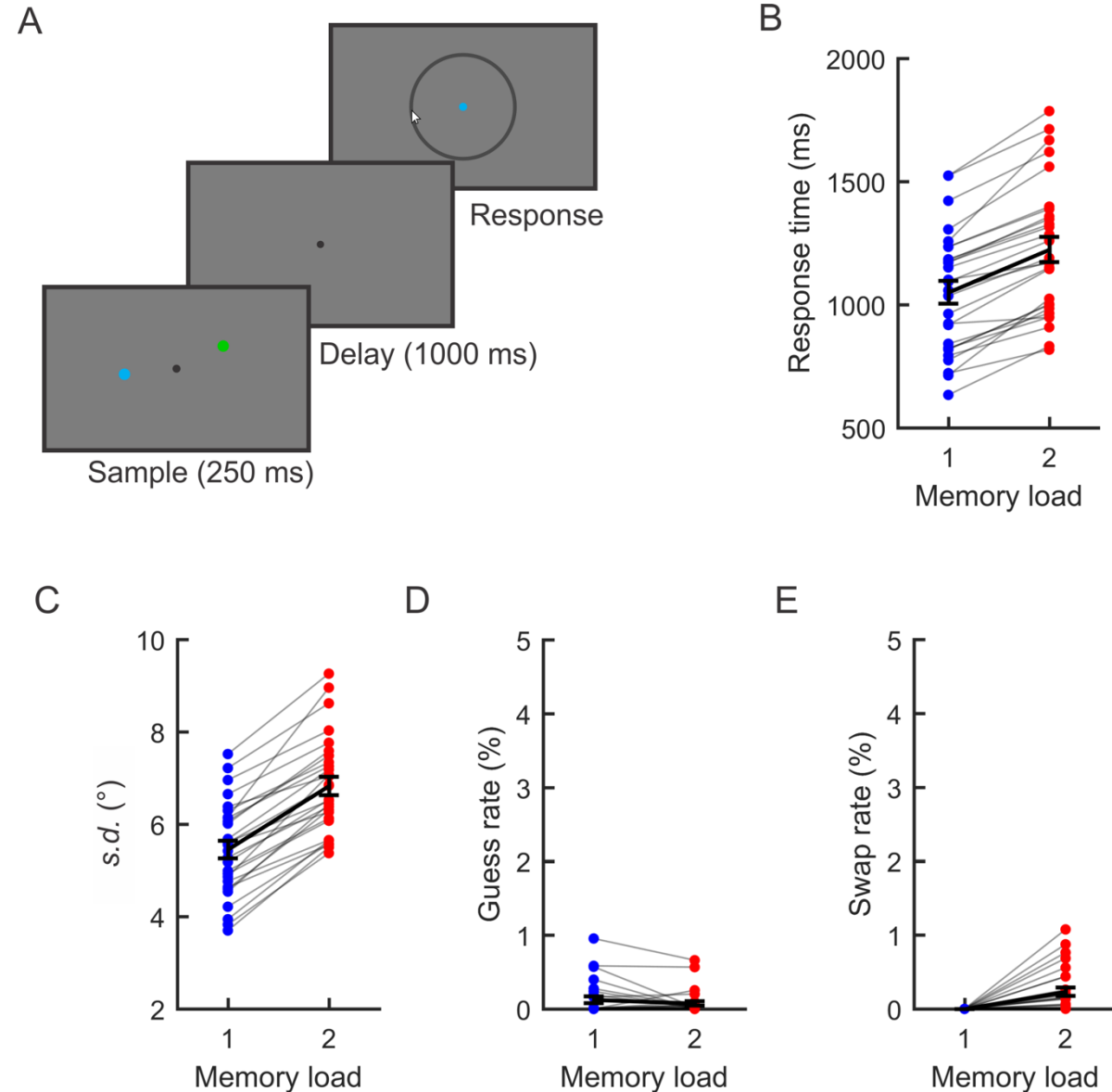
# Attentional control in WM under multiple loads

Sutterer et al.,  
2019

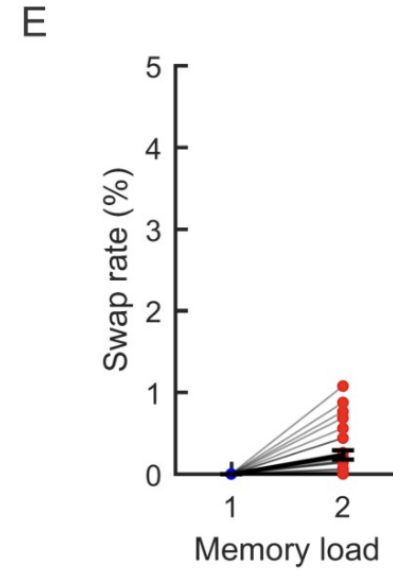
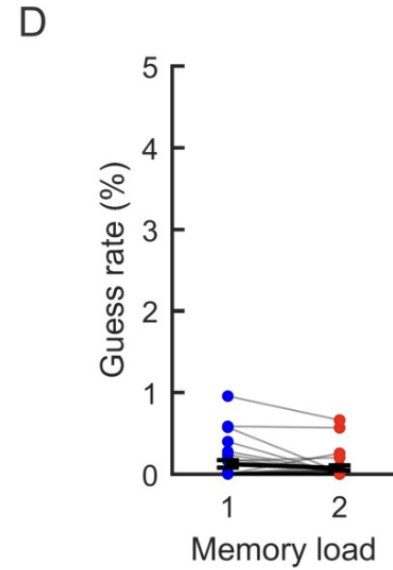
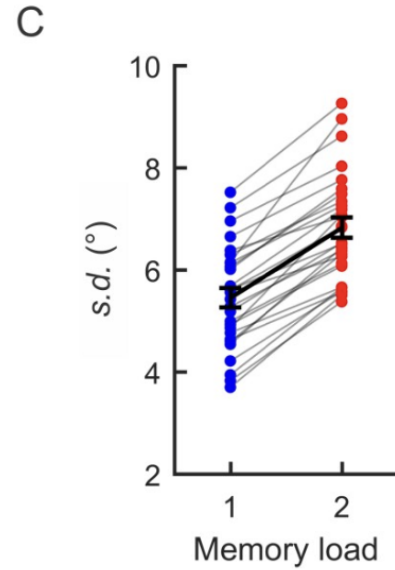
**SD (Memory  
Precision) drops ↓**

**Guessing Rate same**

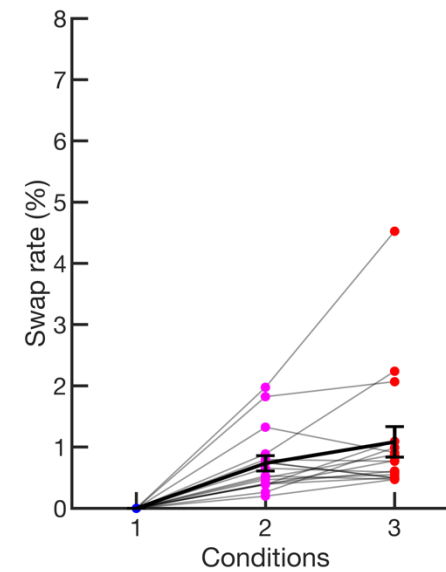
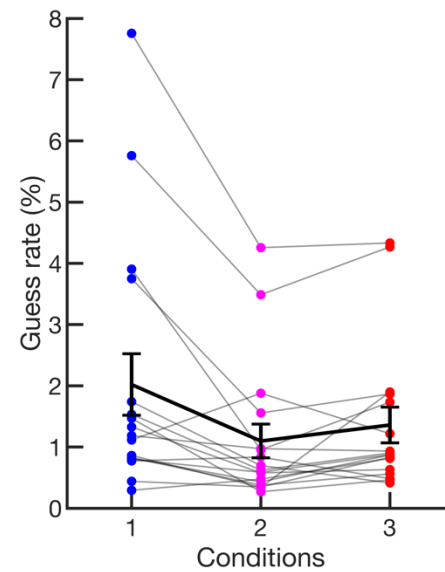
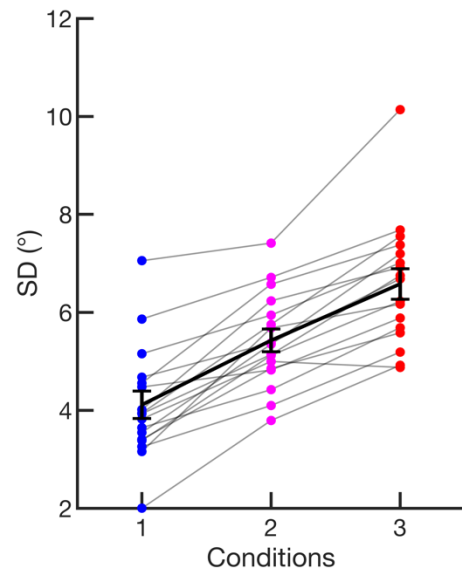
**Swap Rate negligible**



Past  
Works



Ours  
(Exp 1)



# Conclusion: Exp 1

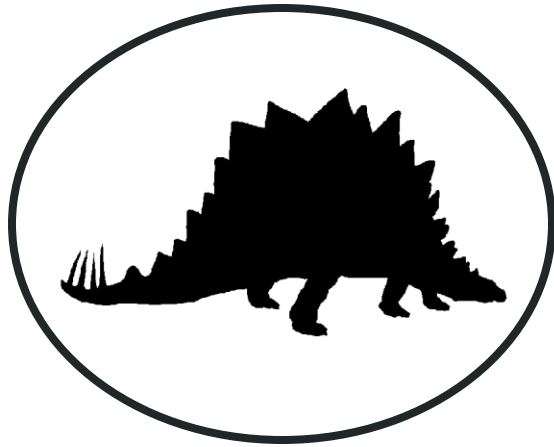
Attention is the neuromechanism that modulated WM, giving possibility that LTM retrieval might have attention as neuromechanism too.

# Experiment 2: Study block

Learning all 162 object-spatial associations one-by-one (eliminate load effect on encoding phase)

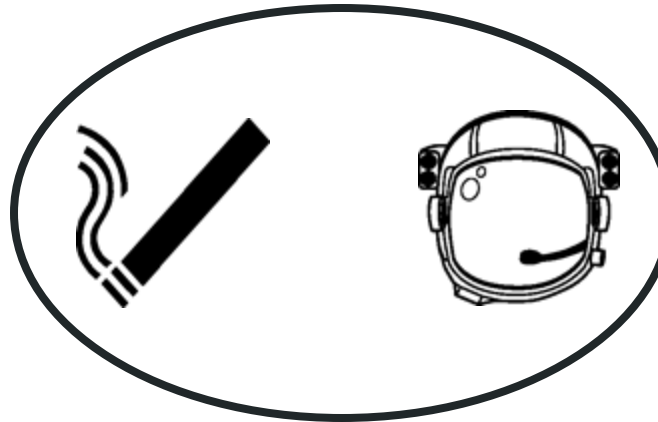


# Retrieval Task Manipulation



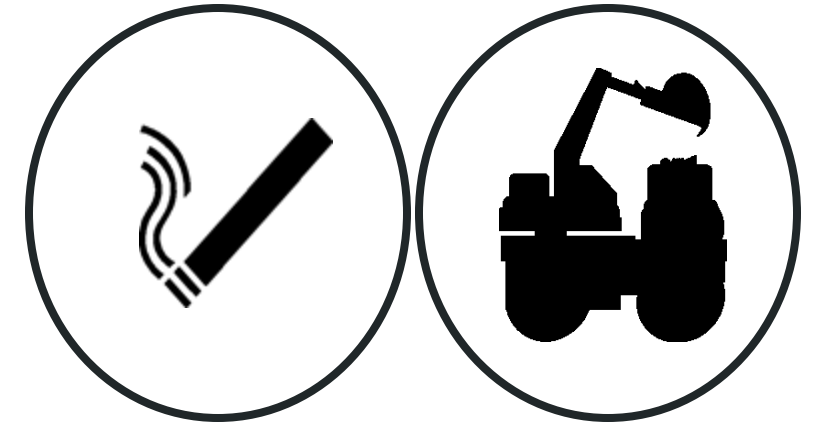
1 recall

- 1 item set size
- 1 attention spot



2 group

- 2 item set size
- 1 attention spots (items latently associated)



2 non-group

- 2 item set size
- 2 attention spots

# Object-spatial associations



There are 27 possible spatial associates on the response ring.

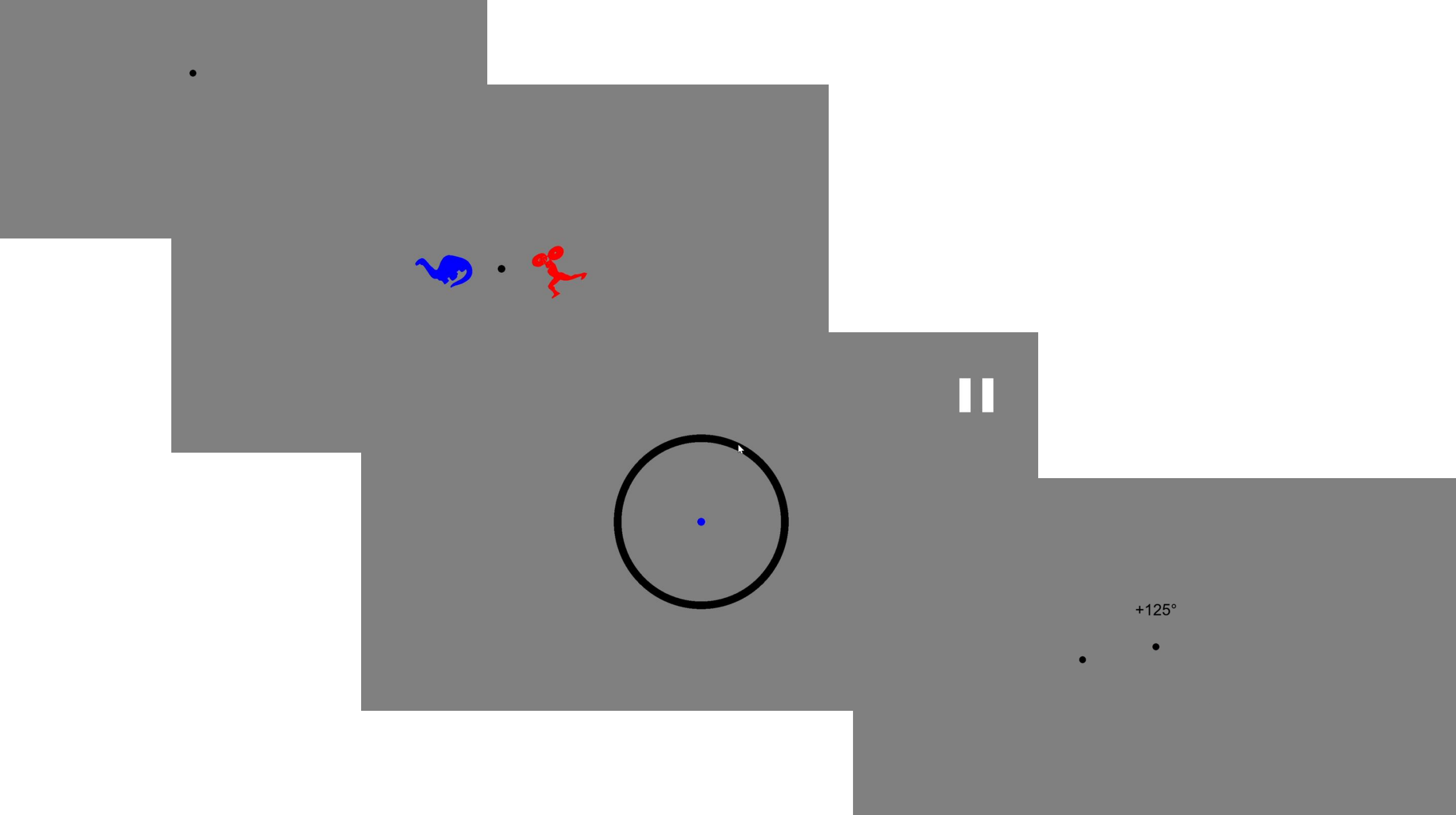
=> 9 equidistant spots (40 degrees apart) x 3 (+ 2 jitters on left and right side of the spot with 10 degrees deviation - scramble mnemonic exploitation of equidistance properties)

27 pairs => Catch trial conditions - on retrieval automatically skip response, help fulfill load in LTM of 1 recall to be comparable with 2 Group and 2 Non-Group conditions

27 pairs => 1 recall

54x2 pairs => 2 Group condition (54 pairs) and 2 Non-Group condition (54 pairs). For 2 Group the object dyads will be present together while 2 Non-Group are randomly permuted

The object will be randomly tilted at  $[-45^\circ \ 45^\circ]$  when being presented to scramble orientation feature exploitation.



# Behavioral Result

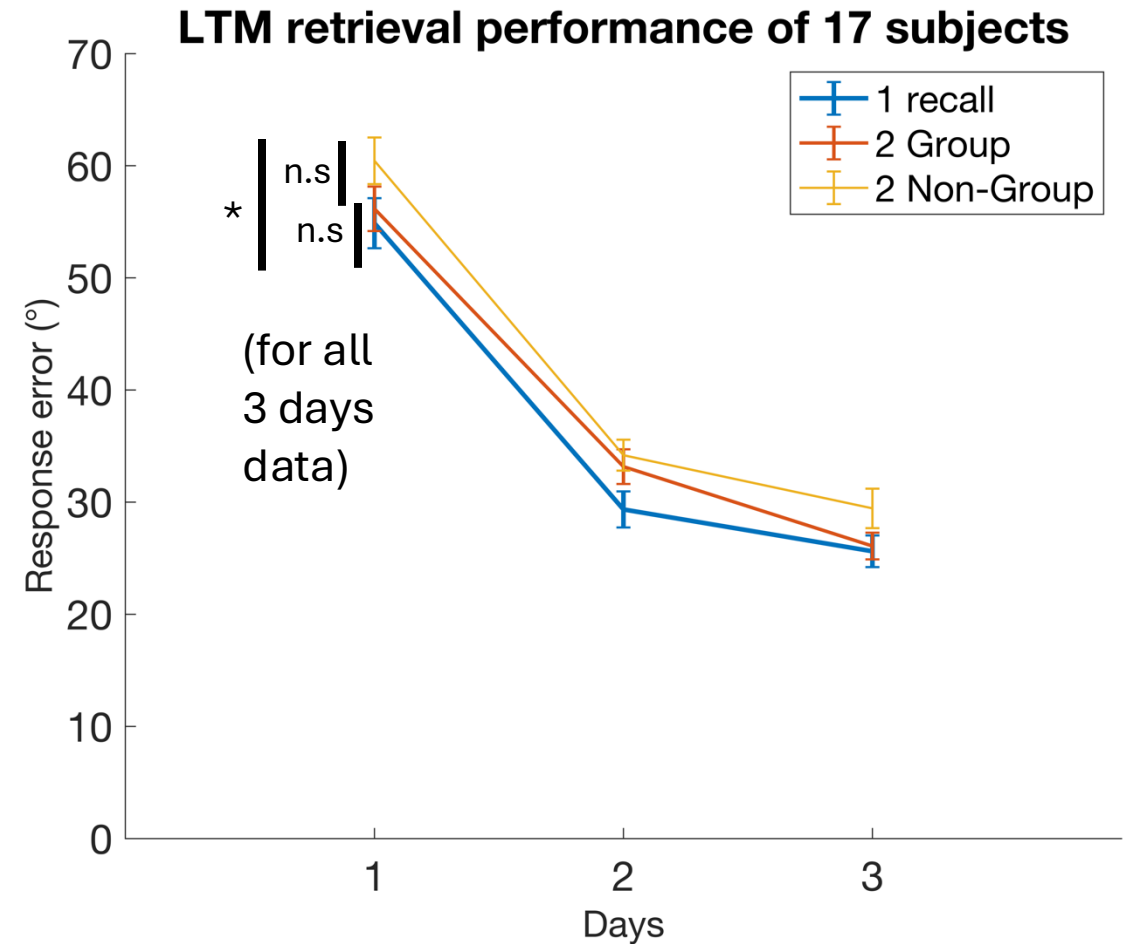
3-ANOVA (Days x LTMcond {1,2g,2ng} x Subject)

- Significant main effect on LTMcond ( $p=0.0011^{**}$ )

- Significant main effect on Days ( $p=0$ )  
=> Have learning effect

2 group is not significantly different from 1 recall and 2 non-group

**=> Is attention not the neuromechanism that LTM retrieval use?**



\*errorbar represents  $\pm 1$ SE

# Attentional control in WM under multiple loads

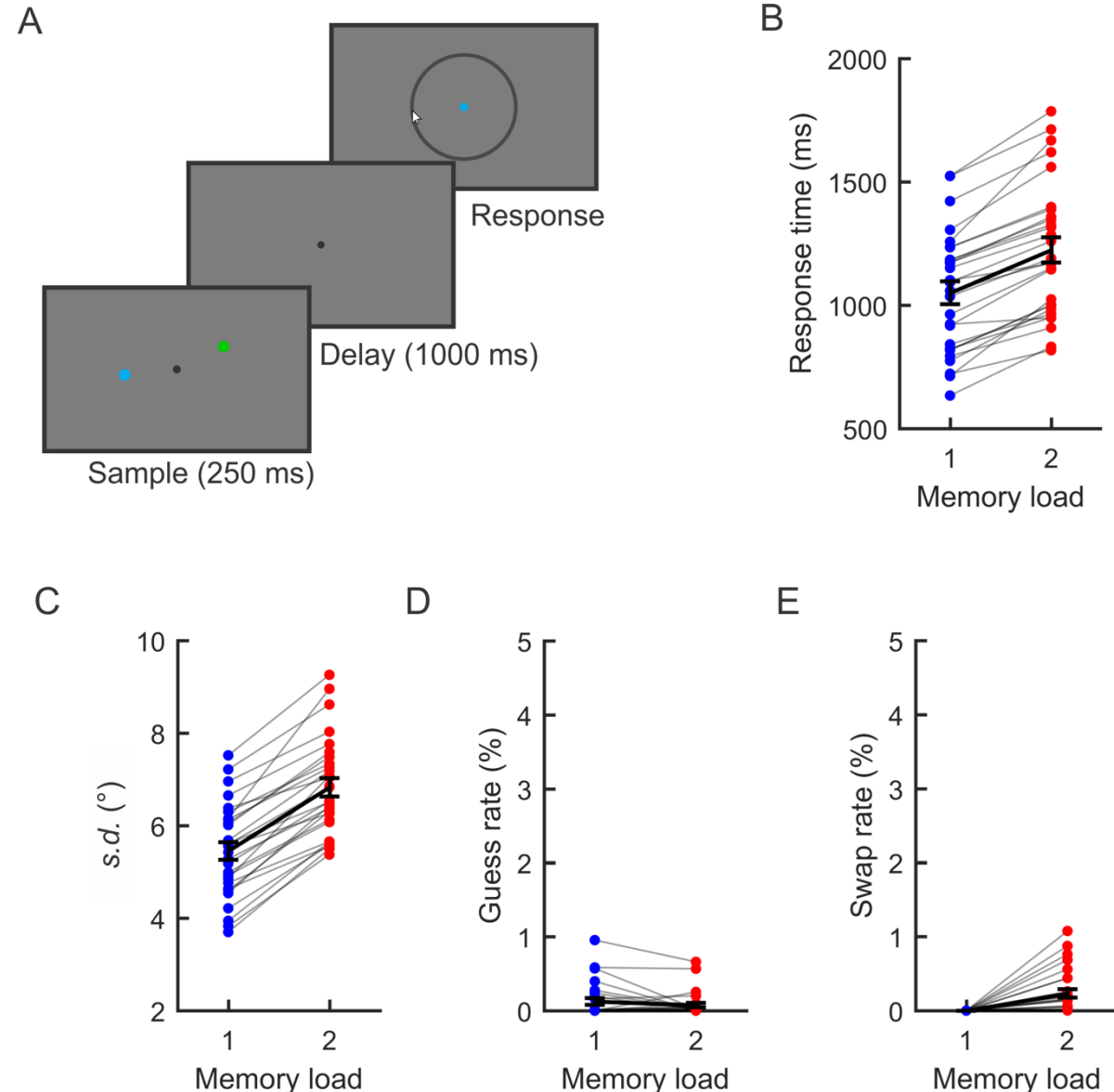
Sutterer et al.,  
2019

**SD (Memory  
Precision)** drops ↓

**Guessing Rate** same

**Swap Rate** negligible

If LTM retrieval use  
attention, then it  
should have the same  
parameter trends?



# Hypothesis

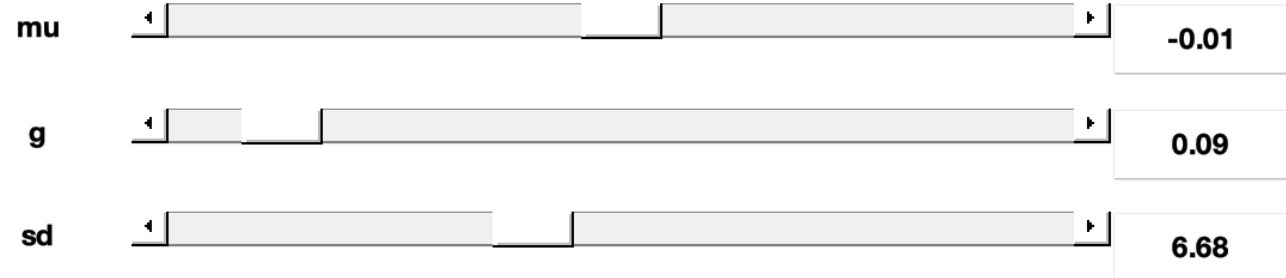
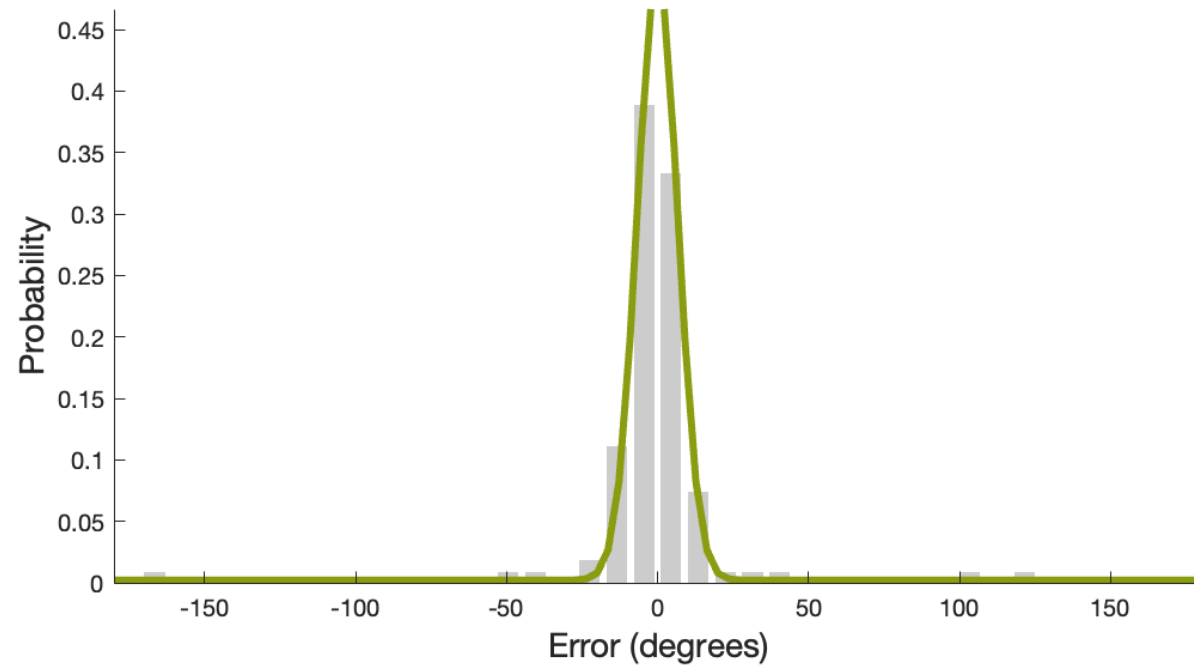
1. WM shared attention mechanism with LTM retrieval process

=> LTM retrieval precision (SD) should **drops ↓** and guessing rate (g) should stays the same on different set size.

• 2. (Alternative) LTM retrieval use decision-making mechanism instead of attention

=> Reversal of trends (SD same but g **drops ↓**)

# MemToolbox Fitting

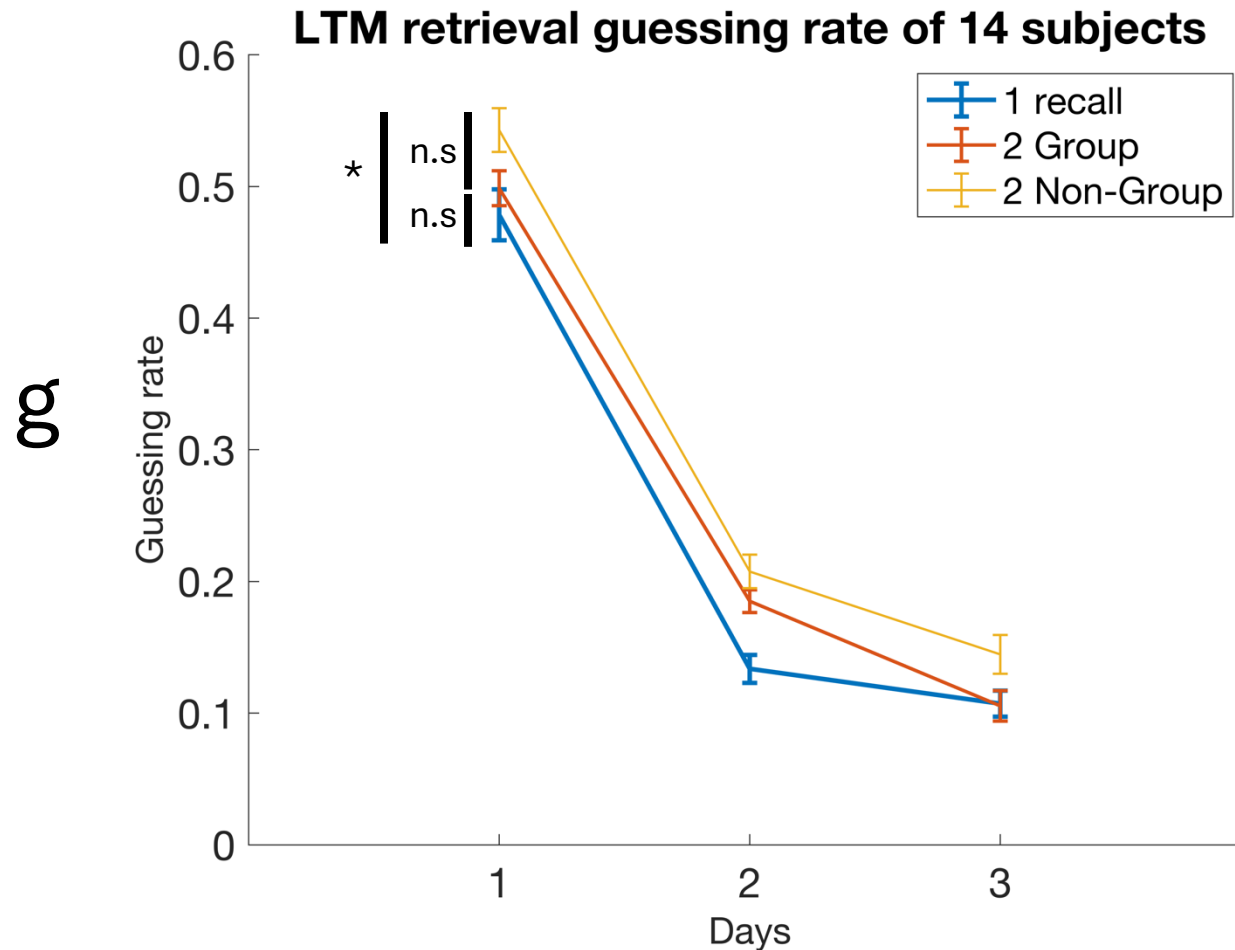


# Model Selection

| Metrics / Model            | Mixture (g, SD) | Swap (g, B, SD) |
|----------------------------|-----------------|-----------------|
| AICc (Corrected Akaike)    | 1020.8 <        | 1022.3          |
| BIC (Bayesian Info. Crit.) | 1034.1 <        | 1040.1          |

Both average AICc and BIC scores (over n=14, 3 days, and 3 LTMcond) in favors of mixture model over swap model (Mixture model has lower score - parsimonious).

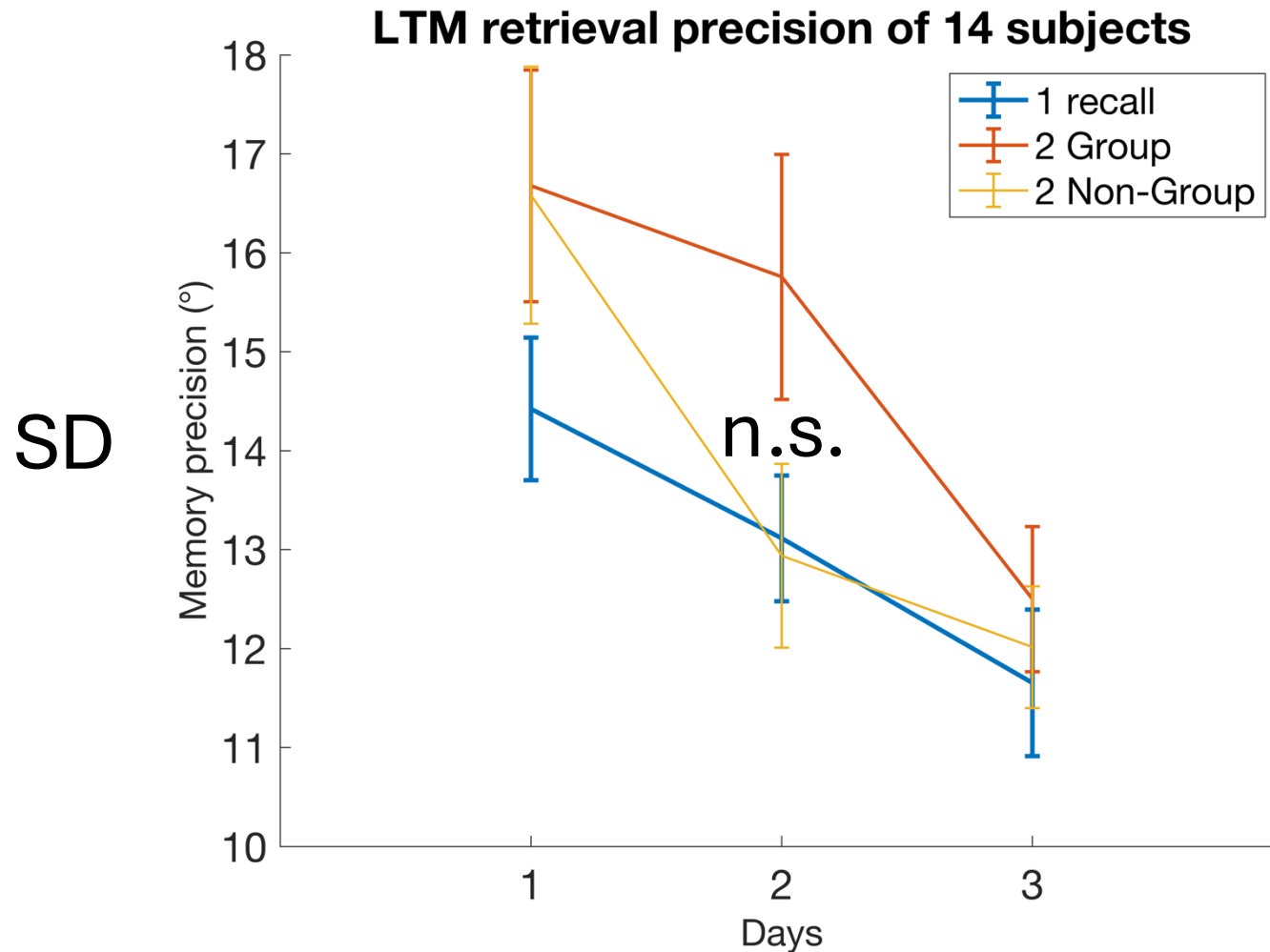
# Mixture Modeling Result



3-ANOVA (Days x LTMcond  
{1,2g,2ng} x Subject)

- Significant main effect on LTMcond ( $p=0.0092^{**}$ )  
=> g **drops** ↓ on less set size!
- Significant main effect on Days ( $p=0$ ) => Have learning effect (Graph is descending)

# Mixture Modeling Result



3-ANOVA (Days x LTMcond  
{1,2g,2ng} x Subject)

- **Non**-significant main effect on LTMcond ( $p=0.3527$ )  
=> SD stays the same on less set size!

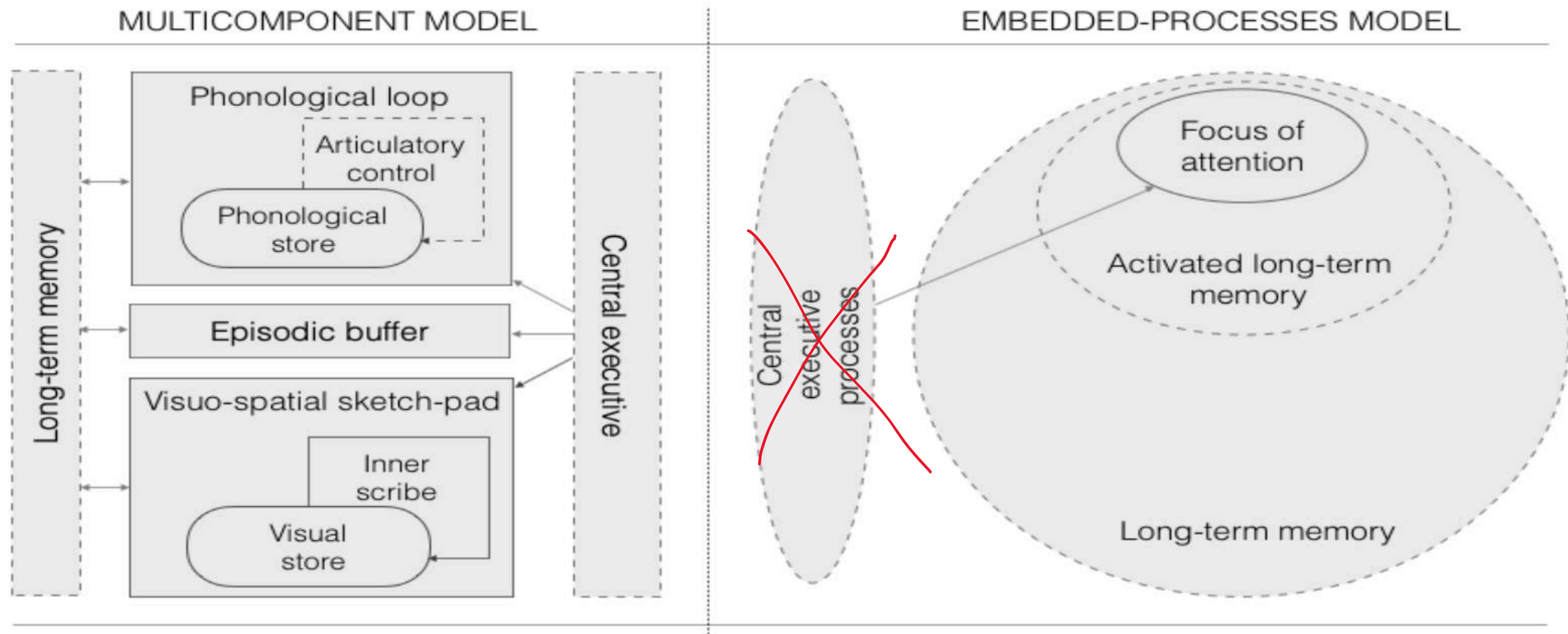
- Significant main effect on Days ( $p=0$ ) => Have learning effect (Graph is descending)

# Result supported alternative hypothesis

LTM retrieval use different mechanism that may involve decision-making rather than attention  
(hence the change in guessing rate)

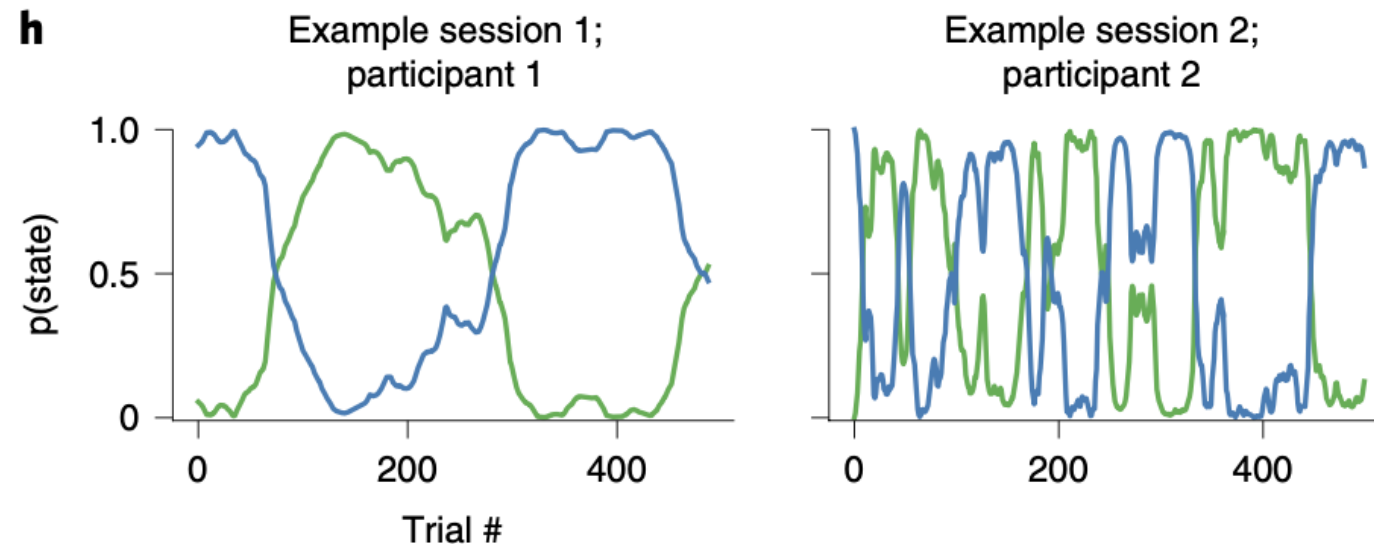
# Implication

- Possible reconciliation of theory with decision-making component!



# Limitation and Future Works

Resolve temporal dynamics on decision-making process during LTM retrieval with GLM-HMM modeling...



# State Space Modeling (GLM-HMM): Experiment 2

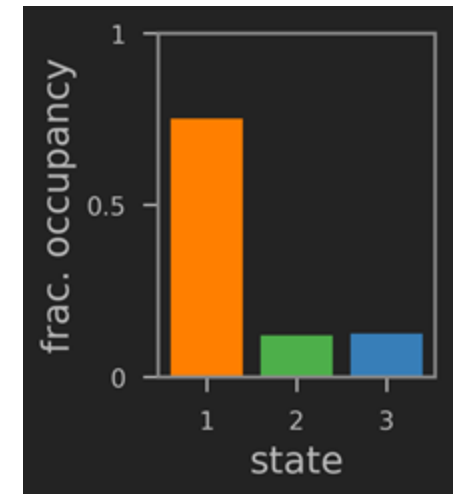
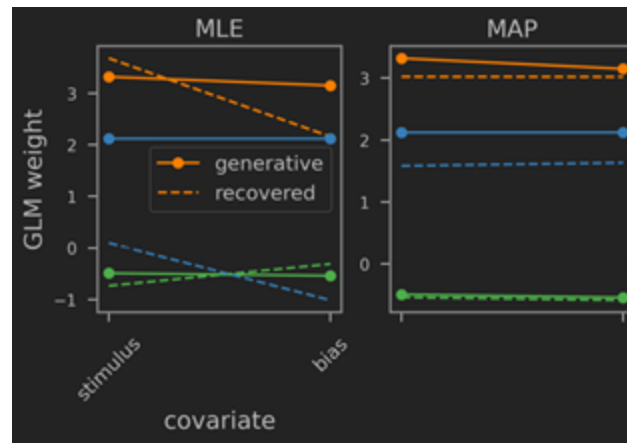
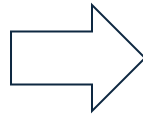
Ashwood et al. (2020) develops Generalized Linear Model-Hidden Markov Model (GLM-HMM) modeling to reveal hidden decision-making strategy states in behavioral datasets.

Generalized Linear Model => The weights represents decision-making states in which multiple GLMs may have different biases towards input (Feature Importance).

Hidden Markov Model => Use to modeled state transitions across trials, no memory property making sure that only previous decision are influential to present decision-making (like current LTM capacity)!

Input (Spatial associates locations, Previous response error etc.)

Observation (2AFC Decision - [0 1] vector)



# Converting LTMdivide to 2AFC

On day 1, there is less to none knowledge in LTM so decision making strategy with biases might come into play here... (Random guessing, item specific variations).

